

Total No. of Questions : 10]

SEAT No. :

P-9338

[Total No. of Pages : 4

[6181]-573

B.E. (Mechanical)

CAD/CAM & AUTOMATION

(2015 Pattern) (Semester - I) (402042)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Use of scientific calculator allowed.
- 5) Assume suitable data, if necessary.

- Q1)** a) With the help of neat sketch, explain various coordinate systems used in computer graphics. [6]
- b) Write homogeneous transformation matrix for translation of triangle ABC through distance t_x and t_y in X and Y direction respectively, where $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$. [4]

OR

- Q2)** The end points of line AB are $A(1, 3, 5)$ and $B(4, 5, 1)$ and line PQ are $P(0, 4, -3)$ and $Q(1, 7, 0)$. Determine :

- a) Parametric equation of line AB & PQ [4]
- b) The unit vectors of line AB & PQ [2]
- c) Check, lines AB & PQ are parallel or not? [2]
- d) Check, lines AB & PQ are perpendicular or not? [2]

- Q3)** a) Compare B-rep & CSG techniques in solid modeling. [5]
- b) Derive element stiffness matrix for two noded 1D bar element. [5]

OR

P.T.O.

- Q4)** An axial stepped bar as shown in Figure 1, is axially loaded by force $P = 50\text{kN}$. Determine nodal displacement, reaction forces and stress in each element.

Assume :

[10]

- $E_1 = 200\text{kN/mm}^2$, $A_1 = 400\text{mm}^2$ and $L_1 = 250\text{mm}$
- $E_2 = 150\text{kN/mm}^2$, $A_2 = 300\text{mm}^2$ and $L_2 = 250\text{mm}$
- $E_3 = 200\text{kN/mm}^2$, $A_3 = 200\text{mm}^2$ and $L_3 = 250\text{mm}$

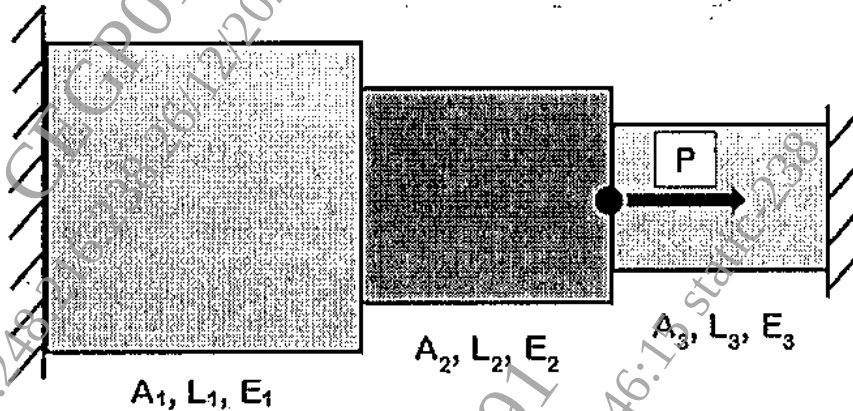


Figure 1. Q. 4

- Q5)** a) Explain Absolute & Incremental positioning in CNC machines. [4]
 b) Write CNC part programming for machining slot with 8 mm diameter end milling cutter as shown in figure 2 in MS plate of $100\text{mm} \times 100\text{mm} \times 25\text{mm}$ thick plate. Assume suitable cutting data. [12]

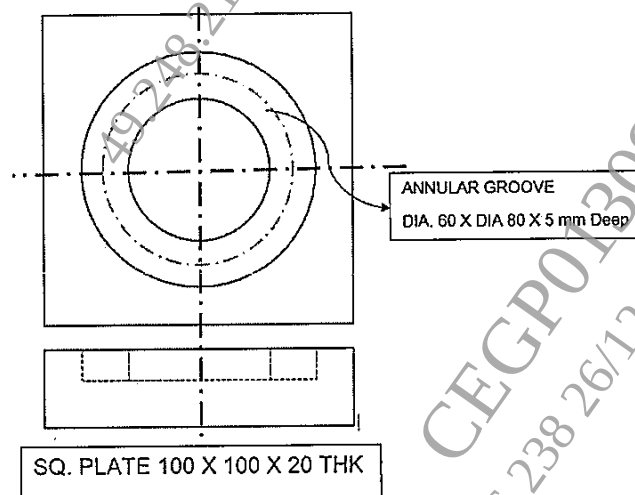


Figure 2. Q. 5(b)

OR

Q6) a) Write G-codes for Cutter Compensation with suitable sketch. [4]

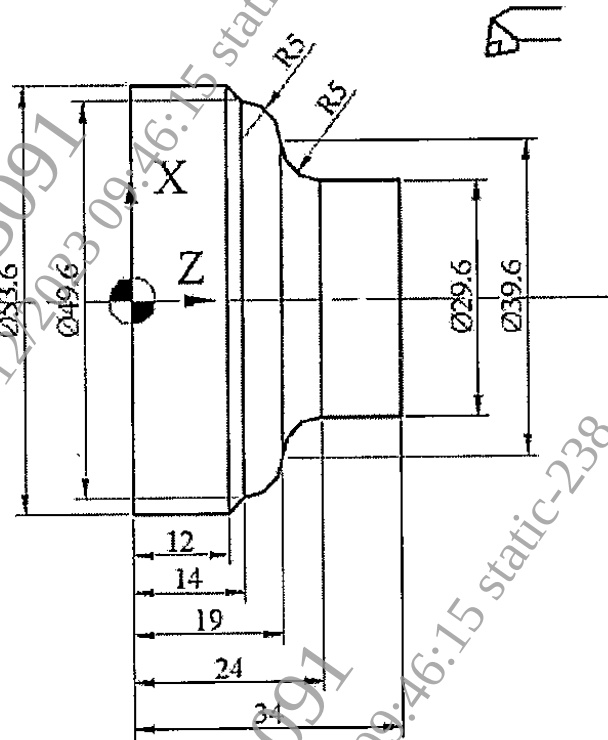


Figure 3. Q.6(a)

b) Write CNC part program for turned component as shown in figure 3 using canned cycles. Assume suitable cutting data. [12]

Q7) a) Explain Fused Deposition Modeling with the help of neat sketch. [8]

b) Discuss need and various components of Product Lifecycle Management (PLM) in detail. [8]

OR

Q8) a) Classify Rapid Prototyping based on working principle. Discuss merits and demerits of RP process. [8]

b) Discuss STL file format in detail. [8]

Q9) a) Discuss Automation in brief. Compare hard, soft and flexible automation with suitable examples. [10]

b) State RIA definition of robotics. State three laws of Robotics. [8]

OR

Q10) Write Short note on any three :

[18]

- a) Group Technology
- b) AGV's
- c) AS/RS systems
- d) PTP Robot
- e) Lead through programming
